Gleipnir: Frontier-Quality AI Control Monitors at Small-Model Cost

Two-week sprint proposal | Jeremias Ferrao | MATS exploration phase | 19 October 2026

1 The problem: synchronous monitoring with a minimal safety tax

The strongest AI control protocols are *synchronous*: a trusted monitor scores each action an untrusted agent proposes *before* it executes, so a suspicious step can be blocked, resampled or escalated before it does harm. This puts the monitor on the critical path of every agent step for every user. Its latency is added to each action and its compute scales with the whole deployment. Frontier models are good monitors but too slow and expensive to run on every step, so in practice one either monitors asynchronously (after the damage) or pays a large *safety tax*. A small monitor with frontier-level quality removes that trade-off: one GPU serves many concurrent agents, and the per-action overhead is a small fraction of the agent's own inference cost.

Gleipnir trains small open-weight monitors for scheming, sabotage, deception and other control-relevant behaviour in agent trajectories. The core recipe is *binary-logit distillation*: a strong teacher's probability on the decision token (`0` | `1`) becomes a soft target for a LoRA student that emits that single token. With no generated rationale, a monitor call is one prefill; Gleipnir 4B scores a ~4K-token trajectory in a median 0.15 s on one B200 in unoptimised BF16. On the out-of-distribution suite of Sinha et al. (never used for selection), Gleipnir 4B is the cheapest point on the cost–performance frontier (cost is the standard proxy, as latency varies widely across providers), and Gleipnir 27B surpasses its own teacher at 1/14 of the cost. **The sprint's goal is to move the 4B towards the 27B's quality at the 4B's cost.**

Monitor	pAUROC@20	$/1K evals
Gleipnir 4B (ours)	**0.782**	**0.26**
Qwen3.5-4B SFT+RL[a]	0.684	0.36
Jev 1.13	0.850	0.39
GPT-6 Luna[b]	0.837	0.84
Gleipnir 27B (ours)	**0.935**	**1.85**
Qwen3.5-27B SFT+RL[a]	0.831	2.78
Kimi K3 (our teacher)	0.908	25.12
Claude Opus 4.6, prompted[a]	0.942	94.79

Mean pAUROC@20 over six OOD sources (6,395 trajectories); uncached list price.
[a] Sinha et al.'s reported result. [b] Binary log-probs, reasoning off.

2 Sprint plan (deliverable: Gleipnir v1 model release + technical report)

Week 1: make the 4B much better, using Gleipnir 27B as an intermediate teacher.

- **Teacher-assistant distillation** (Kimi K3 → Gleipnir 27B → 4B). The 27B shares the 4B's exact 248K-token vocabulary and model family, is now a stronger monitor than Kimi, and is cheap enough to relabel freely. The unadapted 27B was a *worse* teacher than Kimi (−1.9 ID points), so the gain must come from the trained intermediate.
- **Richer supervision channels.** (i) Full-vocabulary KL at the decision position. A first probe diagnostic finds the non-decision tail adds little over the binary margin (+0.04 ID / +0.7 OOD points), so this gets one controlled arm rather than a sweep. (ii) Soft auxiliary targets such as the *index of the harmful action*, giving the student grounded evidence about *where* the misbehaviour is, which also helps human auditors.
- Recipe and checkpoint choices are frozen on the held-out ID set; OOD is read once, at the end.

Week 2: find the blind spots and evaluate on benchmarks the project has never touched.

- **Adversarial attacks on the final monitors.** Because the score is a single logit margin, white-box attacks are straightforward: gradient-guided token optimisation (GCG-style) restricted to text an agent controls (its messages, comments and tool arguments), with embedding-space attacks as an upper bound. Then test whether the attacks transfer to black-box settings, to the 27B and to frontier monitors. Report attack success against edit budget and categorise the blind spots found.
- **Fresh benchmarks.** One-shot evaluation of the frozen release checkpoints on MonitoringBench and SleightBench (unseen throughout the project), with metrics and baselines fixed before scoring.
- **Release.** Model cards, weights, evaluation code and the technical report.

3 Where Neel's input would help most

- **Attack threat model.** White-box gradient attacks are the easiest to run, but a scheming agent will not have gradients. Which attacker (white-box, transfer, LLM-rewriter, latent) would give results the control community finds convincing, and is a white-box blind-spot map still valuable as a worst-case bound?
- **Convincing benchmarks.** Beyond MonitoringBench and SleightBench, which held-out monitoring evaluations would the community find most convincing for a performance–cost claim?
- **Interpretability and scope.** Which mechanistic analyses of the monitor (e.g. what drives its suspicion score) would most strengthen the report, and what to cut if two weeks proves tight.

4 Completed earlier in the exploration phase (included in the final report and release)

- **Inference optimisation.** Prefill-only B200 serving with FP4/MXFP8 kernels: up to 1.6× throughput for the 4B and 2.2–2.3× for the 27B across 4K–1M-token contexts over BF16. When a growing trajectory is re-scored after every action, prefix caching makes monitoring a 1M-token agent session 12.9× faster than re-processing the full history each time.
- **Quantisation-aware distillation (QAD)** so that low-precision deployment copies keep BF16 ranking quality.
- **Serving-time ablation of injection-triggered suspicion.** Monitors flag harmless code once monitor-directed injection or jailbreak text is present. Removing the responsible residual-stream direction at inference time cuts these false alarms to below 0.5%, with no retraining.